Group Assignment -2, Problem 1

Setup: E depends only on A and L, A depends on W and O, L depends on O, and O depends on D.

So,

$$P(E = High) = \sum_{a,l} P(E = High \mid a, l) \cdot P(a, l)$$

where $P(a,l) = \sum_{o} P(o) \cdot P(a \mid o) \cdot P(l \mid o)$

and $P(a \mid o) = \sum_{w} P(w) \cdot P(a \mid w, o)$

Step -1: Occupancy (sum out D)

$P(O = High) = P(weekday) \cdot P(High \mid weekday) + P(weekend) \cdot P(High \mid weekend)$

$= 0.7 \times 0.45 + 0.3 \times 0.8$

$= 0.555$ //

$P(O = Low) = 1 - P(O = High) = 0.445$

Step -2: AC given Occupancy (Sum out W)

$P(A = on \mid O = high) = 0.5(0.9) + 0.3(0.4) + 0.2(0.1)$

$= 0.59$

$P(A = off \mid O = high) = 0.41$

$P(A = on \mid O = low) = 0.5(0.6) + 0.3(0.15) + 0.2(0.05)$

$= 0.355$

$P(A = off \mid O = low) = 0.645$

Step-3: Joint P(A, L) (sum out O)

$P(on, heavy) = 0.555(0.59)(0.75) + 0.445(0.355)(0.35)$

$= 0.300879$

$P(on, light) = 0.555(0.59)(0.25) + 0.445(0.355)(0.65)$

$= 0.184546$

$P(off, heavy) = 0.555(0.41)(0.75) + 0.445(0.645)(0.35)$

$= 0.271121$

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$P(\text{off}, \text{light}) = 0.555\,(0.41)(0.25) + 0.445(0.645)(0.65)$

$= 0.243454$

Step -4 : Energy

$P(E = \text{high}) = 0.92\,(0.300879) + 0.68(0.184546) + 0.55\,(0.271121) + 0.12\,(0.243454)$

$= 0.5806$ ✓

Matches with code output